Juliette Bruce's Scholarship, Teaching, Service

Scholarship. My research is in the intersection of algebraic geometry and commutative algebra, which both focus on studying the duality between systems of polynomial equations and the geometry of their solution sets. As described i my future research plan, my work follows three connected threads.

Syzygies My doctoral work studied the relations among systems of polynomial equations, often called syzygies, that describe higher dimensional shapes. The equations defining a shape are rarely independent, but instead often satisfy relations, and the relations satisfy further relations, and so on. Homological algebra provides us tools to organize this cascade of relations, and understand what it tells us about the geometry of the underly shapes. In my work on asymptotic syzygies, I studied how these syzygies change when we describe the same shape using increasingly complicated coordinate. Breakthrough work of Green and Ein and Lazarsfeld showed that as the degree of the coordinates goes to infinite curves have relatively few syzygies (approaching 0% in an appropriate sense) and higher dimensional varieites have a lot oc syzygies (approach 100% in the limit). My work, showed that if one allowed more complicated coordinates, where different sets of coordinates go to infinite at different rates, one sees intermediate behavior (approaching percentages between 0% and 100%). Such phenomena do not occur for curves, and serve to highlight that connections between syzygies and geometry in higher dimensions. This work resulted in publications in the *Journal of Algebra*, *Journal of Commutative Algebra* showing that the number of syzygies is not always normally distributed, and *Experimental mathematics* using high-throughput computing to generate new examples of syzygies for surfaces together with conjectures. Together this work has laid out a broad foundation for syzygies of higher-dimensional solution sets.

Regularity Regularity is a numerical measure of complexity that simultaneously captures the complexity of the geometry of a shapre and the complexity of the equations defining it. Classically, it is a single number, however, together with Lauren Cranton Heller and Mahrud Sayrafi, I have studied a generalization of the measure where the single number becomes an infinite staircase type region. We have shown that in this generalized setting regularity reaintins some of its nice properties from the classical settings, for example, it can be determined algebraically, but also has a number of pathologies: it is not determined by the syzygies alone, and the staircase can have infinitely many corners. Although in the later case collaborators and I also proved that these pathologies do not arrise when studying relatively nice geometric shapes. Using these we also proved that as one takes powers of the defining polynomial equations the regularity staircases exhibit surprising uniformity, in that they translate in a fixed diraction at a linear rate in terms of the powers you are taking. This work has resulted in one publication in *Advances in Mathematics* and two pre-prints under-review at journals or roughly the same quality. Combined with my work on syzygies fits together to create a unified research program seeking to extend the classical homological algebra tools that developed in the 19th and 20th century to study projective spaces, into tools adapted to studying more complicated geometry spaces.

Tropical Geometry of Moduli Spaces Tropical geometry seeks to study moduli spaces – geometric spaces whose points package together families of solutions to polynomial equations – using the combinatorics of polyhedra, graphs, and more complicated objects like matroids and cell complexes. Using these methods with a number of collaborators I have studied the geometry of moduli spaces that arise naturally in number theory: the moduli space of abelian varieties and Shimura varieties. Doing so we have constructed new cohomology classes, which can be thought of as measuring the number of holes in a shape, and developed new combinatorial tools for understanding these space. For the moduli space of abelian varieties, these classes answered a decade old question of Gruskesky about the existence of cohomology classes in odd-degrees. Further, our work highlighted connects between algebraic geometry and a seeming unrelated area of mathematics, the study of arithmetic groups of matrices. The resulting connections has seen substnatial interest in recent years. This work was highlighted in a plenary lecture by Melody Chan at the 2025 Summer Research Institute, an every 10 year conference highlight some of the best work from the preceeding decade. These efforts have resulted in one publication in *Geometry & Topology*, and three preprints under-review at journals.

Dissemination Since joining Dartmouth in Fall 2024 I have posted five new pre-prints, all of which are under-review at peer-reviewed journals. My paper characterizing multigraded regularity was accepted in *Advances in Mathematics*. Since June 2024 I have given 12 invited research talks at seminars, conferences, and

conferences. have attended six invitation-only research workshops since 2024 including two at Oberwolfach, the Banff International Research Station, the Fields Institute, and the American Institute of Mathematics.

Teaching and Mentoring. My first experiences teaching were as an undergraduate course assistant in inquiry-based courses at the University ofM Michigan. In these courses the students worked together to build and discover mathematics for themselves through carefully curated mix of group work, lecture, and carefully chosen problems. That experience still shapes much of my teaching. I want students to have opportunities to discover and work out an idea, explain it to someone else, and develop their own questions

At Dartmouth, I have taught introductory calculus, undergraduate abstract algebra, and an array of graduate courses. At the undergraduate level most of my teaching focus has been on refining and improving Math 8: Calculus of One and Several Variables. This course serves as the first mathematics course many Dartmouth students take, and at least from the students in my first class, has a somewhat intimidating reputation. I have spent significant time making this course a point of support for students, where they are challenged to meet high academic standard in a supportive enviorment.

The advanced courses I have taught (Math 101, 121, 125) are where my teaching and research meet most directly. Math 125 was a topics course using toric varieties as a touchstone to provide students with grounded introduction to algebraic geometry. Math 121 was a commutative algebra course, which I designed around a series of sixteen inquiry-based worksheets, that build on work of Karen Smith. These worksheets are available freely on my website, and I hope to slowly develop them into an open-source textbook. Additionally, I have taught Math 150, a graduate reading course, two times; in Fall 2026 I led a group of 10 students postivity in algebraic geometry, and in Winter 2026 I led a group of 15 undergraduate and graduate students learning about one of the jewels of 20th century mathematics: The Weil Conjectures.

Mentoring students is one of the parts of my job I find most rewarding. Currently I m advising two Ph.D. students: Will DeGroot (4th year) and Jacob Lehmann Duke (3rd year). Their projects connect with my own interests in computation and the complexity of equations on toric varieties. I expect that within the year I will become Alex Moon's (2nd year) Ph.D. advisor as I have been helping them prepare for their advancement exam, and advising them on a small research project. Since Summer 2026 I have mentored Amy Q Li, a new post doc in the mathematics department.

Beyond Dartmouth, in summer 2026 I led three undergraduates on a summer search program through *Count Me In* held at the University of Edinburgh. All three of these students are now in the process of applying to graduate school. In Fall 2026 I am one of two lectures for the Preliminary Arizona Winter Schoo, a nine-week online course introducing approximately 300 advanced undergraduate students and early graduate students from around the world to recent advancements in computational algebraic geometry.

Service and Community.

At Dartmouth. In fall 2025 I founded *Unwritten Rules*, a graduate seminar that seeks to make the hidden curriculum and expectation of an academic career explicitly. Since its founding this seminar meets over dinner 2-3 times a term to discuss things like attending conferences, applying for funding, and submitting papers to journals. A side-effect of this has been an increased sense of community among the graduate students. I serve as the faculty co-advisor for Dartmouth's student chapter of the Association for Women in Mathematics. In November 2024 I organized the Algebraic Geometry Northeastern Series (AGNES) and BATMOBILE conferences at Dartmouth, which brought over 150 mathematicians to Dartmouth to discuss recent breakthroughs in algebraic geometry. Currently, I am in the early planning stages of organizing *GEMS of Commutative Algebra*, a conference building research networks for early-career researchers in commutative algebra, which I hope to held at Dartmouth in 2028.

In the profession. I was the inaugural president for *Spectra, the Association for LGBTQ+ Mathematicians*, and led the adoption of bylaws, the creation of a invited lecture series at the Joint Mathematics Meetings, and fundraised over $20,000 in seed funding. I am currently serving on the board of trustees for *Spectra* working to guide the organization to longterm stability and sucess. From 2023 to 2025 I served on SLMath's Committee on Women in Mathematics. Currently I serve on the American Mathematical Society's Committee on Engagement, Participation, and Advancement and the Committee on AMS-Simons Travel grants. I have reviewed grants for the National Science Foundation, refereed for more than a dozen different journals. In my career I have organized 18 conferences and workshops that have brought together over 1000 mathematicians. With co-organzier, I have submitted a proposal to the Fields Institute to organize a thematic semester in combinatorial algebraic geometry.